Instructor Solution

Habib University

Introduction to Probability and Statistics – EE 354/CE 361/MATH 310 – Fall 2025

Instructor – Dr. Aamir Hasan - Section L2

Time = 18 Minutes **Quiz No 1** [20 points]

Note - *Provide all your answers on the answer sheet. Present your final answers and reasoning clearly. Most of the total score for the questions will be awarded on your reasoning. Please refrain from random or incoherent scribbles as they will not be graded.*

Question 1 [10 points]

A 6-sided die is tossed and the number of dots facing up is noted.

(a) Find the probability of the elementary events, if faces with an even number of dots are twice as likely to come up as faces with an odd number.

(b) Find the probability of the events: $A = \{more\ than\ 3\ dots\}$; $B = \{odd\ number\ of\ dots\}$;

(c) Find the probability of $A \cup B$, $A \cap B$, A^c

Question 2 [10 Points]

Two numbers (x, y) are selected at random from the interval $[0, 1]$, under uniform distribution i.e., that is points are equally likely in the unit square.

(a) Find the probability that the pair of numbers are inside the unit circle.

(b) Find the probability that $y > 2x$.

Question 1 (a) Probability of getting 1, 3, 5 $\Rightarrow P_1 = P_3 = P_5 = P$ 1pt

Probability of getting 2, 4, 6 $\Rightarrow P_2 = P_4 = P_6 = 2P$

$\Rightarrow 3p + 6p = 1 \Rightarrow$ $\boxed{P = 1/9 \text{ for } 1, 3, \text{ or } 5 \quad = 2/9 \text{ for } 2, 4, \text{ or } 6}$ 4pts

(b) $A = \{4, 5, 6\}$ dots ; $B = \{1, 3, 5\}$

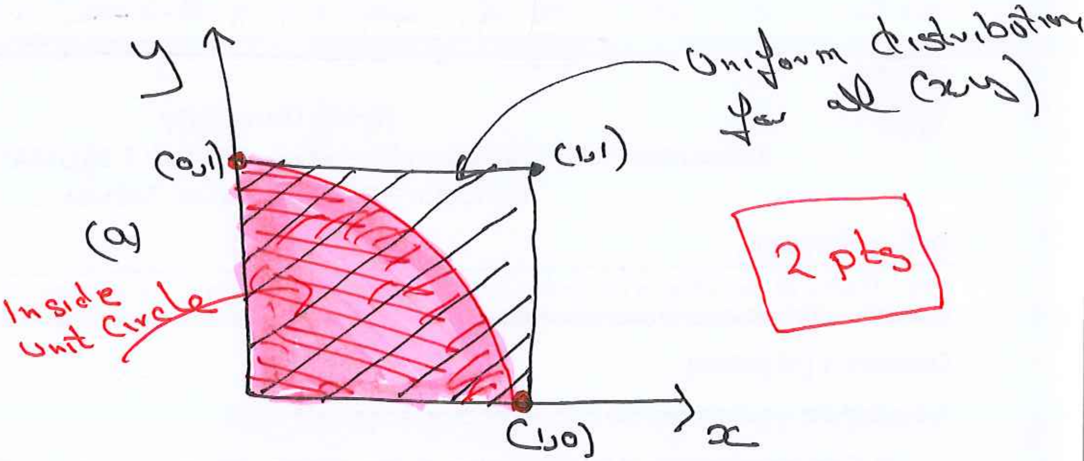
$P(A) = P_4 + P_5 + P_6 = 5p = \boxed{5/9}$ 1pt ; $P(B) = P_1 + P_3 + P_5 = 3p = \boxed{3/9 = 1/3}$ 1pt

(c) 1) $P(A \cup B) = P[\{1, 3, 4, 5, 6\}] = P_1 + P_3 + P_4 + P_5 + P_6 = 3p + 4p = 7p = \boxed{\frac{7}{9}}$ 1pt

2) $P(A \cap B) = P_5 = \boxed{P = 1/9}$ 1pts

3) $P[A^c] = 1 - 5/9 = \boxed{4/9}$ 1pt

Question 2

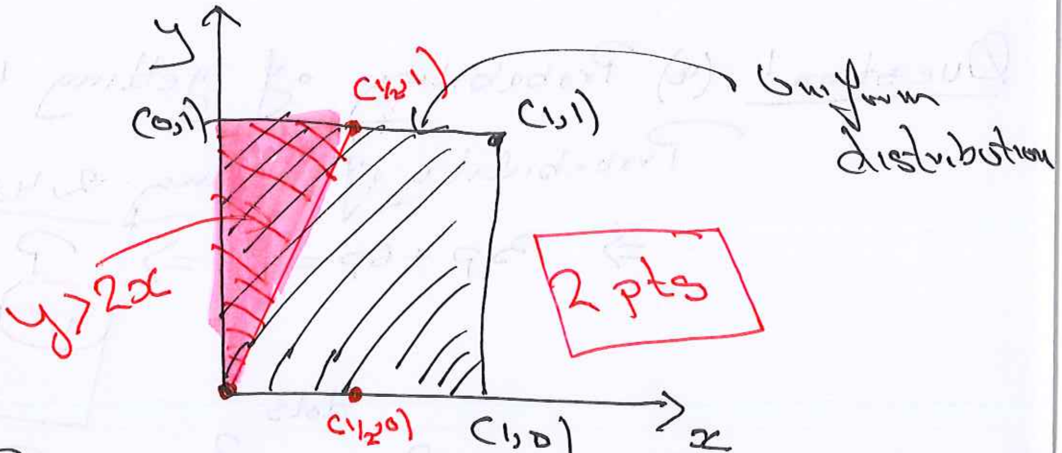


$\boxed{2 \text{ pts}}$

(a) Inside Unit Circle $= P\left[x^2+y^2<1\right]$

$$= \frac{\pi \cdot 1^2}{4} = \frac{\pi}{4} \quad \boxed{3\text{pts}}$$

(b)



$\boxed{2 \text{ pts}}$

$$P\left[y>2x\right] = \frac{1}{2} \times \frac{1}{2} \times 1 = \frac{1}{4} \quad \boxed{3\text{pts}}$$